35 Advanced NumPy & Pandas Practice Questions

Practice set for Data Science learners (advanced level).

#	Question
1	10×10 random NumPy array generate karo aur diagonal elements ka sum find karo.
2	Ek array me top 5 largest values without sorting entire array find karo.
3	1–100 tak array create karo aur multiples of 7 remove karo using vectorization.
4	5×5 array me border elements ko 0 aur inner elements ko 1 set karo.
5	Ek array me duplicate values detect karke unique values return karo without using np.unique().
6	1000 random numbers generate karo aur outliers detect karo using Z-score.
7	2 arrays given hain — common elements find karo without using built■in intersection function.
8	10×10 matrix me row with maximum sum identify karo.
9	Array me second largest element find karo without sorting.
10	Ek array ko normalize karo (0–1 scale) using formula (x-min)/(max-min).
11	Pandas dataframe me department wise top 2 salary employees find karo.
12	DataFrame me salary ka rank column create karo (highest = rank 1).
13	Department wise salary difference between highest and lowest employee find karo.
14	DataFrame me duplicate rows detect karke remove karo but latest entry retain karo.
15	Ek column me missing values hain — unhe department wise mean se fill karo.
16	DataFrame me salary growth percentage column create karo based on previous row salary.
17	Department wise median salary calculate karo without using .median().
18	Employees jinki salary department average se jyada hai unko filter karo.
19	DataFrame me rolling average salary (window size = 3) calculate karo.
20	Ek column me text data hai — sabse frequent word detect karo.
21	Multi-index dataframe create karo aur usme se specific index slice karo.
22	Top 3 departments with highest average salary find karo.
23	Salary column ko salary bands me convert karo (Low, Medium, High).
24	DataFrame ko pivot table me convert karo (department vs avg salary).
25	Employees jinka experience department ke median se jyada hai identify karo.
26	Pandas me custom aggregation function likho jo salary variance calculate kare.
27	Correlation matrix calculate karo aur highest correlated pair identify karo.
28	Dataset me salary outliers detect karo using IQR method.
29	Do dataframes ko merge karo aur mismatched records identify karo.
30	Time-series dataframe me month wise growth rate calculate karo.
31	1 million rows dataframe create karo aur fastest method se groupby aggregation perform karo.
32	Vectorization vs loop performance compare karo NumPy me.
33	DataFrame me lag feature create karo (previous row value).

34	Dataset me customer churn probability calculate karo using conditional filtering.
35	Large dataset me memory optimization techniques apply karo.